Drivers of Out-of-Pocket Health Expenditure: A Fixed-Effects Panel Analysis of 188 Countries, 2000–2024

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Abstract

This paper investigates the macroeconomic and demographic determinants of household out-of-pocket (OOP) health spending using a balanced panel of 188 countries observed annually from 2000 to 2024 (N = 4,700). Drawing on World Bank Development Indicators, I estimate pooled OLS and country fixed-effects (FE) specifications, selecting between them on the basis of a Hausman test. Government health expenditure share emerges as the most economically and statistically significant predictor: a one-percentage-point increase in the public share is associated with a 4.7-percentage-point decline in OOP share within countries over time. GDP per capita and population ageing also matter, while consumer price inflation has no detectable effect. The findings reinforce a well-established cross-country pattern: where the state pays more for health, households pay less.

1. Introduction

Out-of-pocket (OOP) health spending - the share of healthcare costs paid directly by households at the point of use is one of the most regressive forms of health financing. High OOP burdens deter the uptake of needed care, generate catastrophic expenditure for low-income households, and push an estimated tens of millions of people into poverty each year. Reducing OOP exposure is therefore a central plank of the World Health Organization's universal health coverage agenda, embedded in Sustainable Development Goal 3.8.

The empirical literature has produced robust evidence that OOP shares fall as economies develop and as governments take on a larger share of health financing.[1] Yet the magnitude of these effects, and the role of demographic structure and macroeconomic stability, remain contested. Recent work by Kaladharan and Manayath, for example, finds that in emerging economies higher general government health expenditure is associated, somewhat counter-intuitively, with higher OOP spending per capita.[2] The earlier and broader study by Ke, Saksena, and Holly, by contrast, documents the expected negative relationship across a 143-country panel.[3] These differences are partly methodological - emerging-market subsamples versus global panels, OOP measured in levels versus shares, varying control sets and partly substantive, reflecting genuine heterogeneity in how public spending crowds in or out of private payment.

This paper revisits the question on a wide, recent panel. The research question is:

[1] World Health Organization, 'Universal Health Coverage', WHO Fact Sheets (2023), https://www.who.int/news-room/fact-sheets/detail/universal-health-coverage.

[2] Sanju Kaladharan and Dhanya Manayath, 'Out-of-Pocket Healthcare Expenditure in Emerging Economies: Evidence from Panel Data Analysis', Journal of Medical Access, 8 (2024), pp. 1–13, doi:10.1177/27550834241262108.

[3] Xu Ke, Priyanka Saksena, and Alberto Holly, 'The Determinants of Health Expenditure: A Country-Level Panel Data Analysis' (Geneva: World Health Organization / Results for Development Institute, Working Paper, December 2011), pp. 1–28.

What macroeconomic and demographic factors explain cross-country and within-country variation in out-of-pocket health spending between 2000 and 2024?

The central hypothesis is that, holding other factors constant, a larger public share in health financing reduces the household OOP share, a within-country relationship that should survive controls for income, age structure, and inflation. The analysis proceeds in five sections: a brief review of related literature (2), a description of the data and methodology (3), regression results and diagnostics (4), discussion (5), and a short conclusion (6).

2. Related Literature

Cross-country studies of health expenditure typically frame OOP spending as the residual of a national health-financing identity in which public schemes, social insurance, and private pre-payment substitute for direct household payment. Ke, Saksena, and Holly examine this identity in a 143-country panel covering 1995–2008 and find that GDP per capita is the single most powerful predictor of overall health spending, while the OOP share falls systematically with income and with government revenue capacity.[4] A subsequent literature has refined this picture: Sirag and Mohamed Nor identify a non-linear threshold (around 29% of total health expenditure) above which OOP spending becomes poverty-inducing; Kaladharan and Manayath, restricting attention to emerging economies, find that voluntary health insurance and compulsory schemes reduce OOP per capita, but that government health expenditure correlates positively with OOP, a finding they attribute to the simultaneous expansion of both public and private health markets at intermediate income levels.[5]

Three methodological points recur across this literature. First, country fixed effects are essential: unobserved institutional features - the design of health systems, cultural attitudes toward state provision, the maturity of insurance markets vary enormously across countries but slowly within them. Second, OLS standard errors understate uncertainty in panels with persistent serial correlation; clustering at the country level is now standard practice. Third, results are sensitive to whether OOP is measured as a share of current health expenditure (the dependent variable used here) or in per-capita levels. The present study adopts the share specification, which is the more common choice for cross-country welfare comparisons.

3. Data and Methodology

3.1 Data

The data are drawn from the World Bank's World Development Indicators (WDI), accessed in May 2026.[6] The full panel comprises 188 countries observed annually from 2000 to 2024, yielding 4,700 country-year observations. The dependent variable is the OOP share of current health

[4] Ke, Saksena, and Holly, 'Determinants of Health Expenditure', pp. 10–14.

[5] Kaladharan and Manayath, 'Out-of-Pocket Healthcare Expenditure', pp. 4–6.

[6] World Bank, 'World Development Indicators', DataBank (2025), https://databank.worldbank.org/source/world-development-indicators.

expenditure. Four regressors are used: the public share of current health expenditure (gov_health_exp), the natural log of GDP per capita in constant 2015 US dollars (log_gdp_per_capita), the share of the population aged 65 and over (pop_65_plus), and annual consumer price inflation (inflation_rate). Variable definitions are summarised in Table 1.

Table 1. Variable definitions and data sources.

Variable	Definition	Unit	Source
oop_spending	Out-of-pocket expenditure as a share of current health expenditure	%	World Bank WDI
gov_health_exp	Domestic general government health expenditure as a share of current health expenditure	%	World Bank WDI
log_gdp_per_capita	Natural logarithm of GDP per capita (constant 2015 US$)	log USD	World Bank WDI
pop_65_plus	Population aged 65 and above as a share of total population	%	World Bank WDI
inflation_rate	Annual consumer price inflation	%	World Bank WDI

Source: World Bank, World Development Indicators (2025). All variables observed annually, 2000–2024.

Missingness in the raw extract ranged from 0.8% (GDP per capita) to 7.4% (government health expenditure). No country was missing entirely for any variable. To preserve the balanced-panel structure required for fixed-effects estimation, missing values were imputed using a distance-weighted K-nearest-neighbours (KNN) procedure with k = 5.[7] As a sensitivity check, all reported results were re-estimated on the unimputed sample with listwise deletion; the substantive conclusions are unchanged.

Table 2. Descriptive statistics.

Variable	Mean	Std. Dev.	Min	Max	N
oop_spending	35.28	19.43	0.06	87.90	4,700
gov_health_exp	3.04	2.15	0.06	11.80	4,700
log_gdp_per_capita	8.45	1.52	4.70	11.83	4,700
pop_65_plus	8.03	5.82	0.86	29.80	4,700
inflation_rate	8.70	25.36	-16.86	557.20	4,700

Notes: the original gdp_per_capita has been excluded as we already have log_gdp_per_capita.

[7] Olga Troyanskaya et al., 'Missing Value Estimation Methods for DNA Microarrays', Bioinformatics, 17.6 (2001), pp. 520–25. Although developed for genomic data, the KNN imputer is widely applied to economic panels for its ability to preserve cross-sectional structure.

The Hausman test is computed with unclustered standard errors, as required by the test's asymptotic distribution; clustered standard errors are then applied to the chosen FE specification for inference.

3.2 Empirical strategy

The baseline pooled OLS specification is:

$$OOP_{it} = \beta_0 + \beta_1\ GovHealth_{it} + \beta_2\ logGDP_{it} + \beta_3\ Pop65_{it} + \beta_4\ Inflation_{it} + \varepsilon_{it}$$

where i indexes countries (i = 1, …, 188) and t indexes years (t = 2000, …, 2024). Pooled OLS treats every country-year as an independent observation; this is unlikely to be appropriate given the institutional heterogeneity discussed in section - 2. The fixed-effects (within) estimator augments the specification with country-specific intercepts α_i that absorb all time-invariant heterogeneity:

$$OOP_{it} = a_i + \beta_1\ GovHealth_{it} + \beta_2\ logGDP_{it} + \beta_3\ Pop65_{it} + \beta_4\ Inflation_{it} + \varepsilon_{it}$$

Selection between pooled OLS, random-effects, and fixed-effects estimators follows a standard testing sequence: the F-test for poolability is used to evaluate whether country effects matter at all, and the Hausman test compares the consistency of the random-effects and fixed-effects estimators.[8] Inference uses country-clustered standard errors to accommodate the heteroskedasticity flagged by the Breusch–Pagan test and the within-country serial correlation suggested by the Durbin–Watson statistic on pooled-OLS residuals.[9] [10]

[8] Jerry A. Hausman, 'Specification Tests in Econometrics', Econometrica, 46.6 (1978), pp. 1251–71.

[9] Trevor S. Breusch and Adrian R. Pagan, 'A Simple Test for Heteroscedasticity and Random Coefficient Variation', Econometrica, 47.5 (1979), pp. 1287–94.

[10] Halbert White, 'A Heteroskedasticity-Consistent Covariance Matrix Estimator and a Direct Test for Heteroskedasticity', Econometrica, 48.4 (1980), pp. 817–38; A. Colin Cameron and Douglas L. Miller, 'A Practitioner's Guide to Cluster-Robust Inference', Journal of Human Resources, 50.2 (2015), pp. 317–72.

4. Results

4.1 Main estimates

Table 3. Determinants of out-of-pocket health expenditure share.

	Pooled OLS	**Fixed Effects**
	(1)	*(2)*
gov_health_exp	−6.607*** (0.451)	−4.748*** (0.391)
log_gdp_per_capita	−3.168*** (0.711)	−2.262*** (0.718)
pop_65_plus	1.230*** (0.186)	0.232** (0.194)
inflation_rate	−0.0003 (0.025)	0.008 (0.013)
Constant	72.290*** (5.568)	66.926*** (5.4691)
Country fixed effects	No	Yes
Clustered SE (country)	Yes	Yes
Observations	4,700	4,700
Countries	188	188
R^2 (overall / within)	0.458	0.239 (within)
F-statistic	78.22***	353.62***

*Notes: Dependent variable is out-of-pocket spending as a share of current health expenditure (%). Standard errors clustered by country in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Column (2) includes 188 country fixed effects; within-R^2 reported.*

Table 3 reports the two specifications side by side. The coefficient on government health expenditure share is negative, statistically significant at the 1% level, and economically large in both columns. The pooled OLS coefficient of -6.61 implies that a country with a 10-percentage-point higher public share has, on average, a 66-percentage-point lower OOP share - a between-country effect that includes confounded institutional differences. The within-country FE estimate of -4.75 strips out these time-invariant differences and is the more credible causal estimate: holding other factors fixed, when a country raises the public share of health financing by one percentage point, its OOP share falls by roughly 4.75 percentage points in the same year.

GDP per capita enters negatively and significantly in both columns, but its magnitude shrinks from -3.17 (pooled) to -2.26 (FE). The interpretation is conventional: a 1% increase in real GDP per capita is associated with a 0.023-percentage-point reduction in OOP share within a country over time. Population ageing has the expected positive sign in pooled OLS (older populations carry higher OOP burdens, partly because elderly health needs are less likely to be fully covered by public schemes), and the effect shrinks substantially under fixed effects (from 1.23 to 0.23) but remains significant. Inflation is statistically indistinguishable from zero in both specifications, suggesting that household OOP shares respond to structural rather than cyclical macroeconomic forces.

4.2 Specification tests and diagnostics

Table 4. Specification tests and residual diagnostics.

Test	Statistic	Conclusion
VIF (max, excl. constant)	2.40	No multicollinearity
Breusch–Pagan (Pooled OLS)	$p < 0.001$	Heteroskedasticity present
Durbin–Watson (Pooled OLS)	0.29	Severe serial correlation
Hausman test (FE vs RE)	24.76 ($p = 0.0002$)	Fixed effects preferred
F-test for poolability	67.00 ($p < 0.001$)	Pooling rejected
Breusch–Pagan (FE residuals)	184.38 ($p < 0.001$)	Heteroskedasticity remains
Durbin–Watson (FE residuals)	1.12	Positive autocorrelation

Notes: VIF is reported on the pooled-OLS design matrix. Breusch–Pagan and Durbin–Watson statistics are computed on regression residuals.

The diagnostics in Table 4 jointly motivate the FE specification. Variance Inflation Factors for all regressors are below 2.5, well under any conventional threshold for multicollinearity. The Breusch-Pagan statistic decisively rejects homoskedasticity in the pooled-OLS residuals, and the Durbin-Watson statistic of 0.29 indicates severe positive serial correlation - both signs of misspecification consistent with omitted country effects. The Hausman test (statistic = 24.76, $p = 0.0002$) rejects the random-effects null in favour of fixed effects, and an F-test for poolability rejects the restriction that country intercepts are jointly zero at the 1% level. In the FE residuals, heteroskedasticity persists (handled through clustering) and serial correlation is much reduced, with a Durbin-Watson of 1.12. This is an improvement but not a clean bill of health: a dynamic-panel specification with a lagged dependent variable would be a natural next step (see section - 5).

5. Discussion

The headline result that increases in the public share of health financing translate roughly one-for-three into reductions in the OOP share is broadly consistent with the Ke, Saksena, and Holly findings on a 143-country panel from 1995 - 2008, although the implied elasticity here is somewhat smaller.[11] The contrast with Kaladharan and Manayath, who find a positive relationship in their emerging-economies sample, is instructive: their dependent variable is OOP per capita rather than OOP share, and their sample is restricted to middle-income economies in which both public and private health spending tend to rise together as GDP grows.[12] On the share measure used here, and on the wider global panel, the substitution effect dominates the income effect.

Three caveats deserve emphasis. First, the FE specification only identifies the effect from within-country changes; it cannot say anything about why some countries start the period with much higher OOP shares than others, and that between-country variation is where most of the welfare-relevant inequality lies. Second, the Durbin-Watson of 1.12 indicates some remaining within-country serial correlation, which clustered standard errors partially but not fully address. A dynamic specification (Arellano-Bond or system-GMM) with a lagged dependent variable would explicitly model the persistence of OOP shares and is the natural extension. Third, the model is silent on health-system design: a country that shifts from a Bismarckian to a Beveridge model, or that introduces a national insurance scheme, might experience a large discrete change in OOP share that the linear specification cannot capture.

Notwithstanding these caveats, two practical implications are robust to specification choice. Public-share expansion appears to be the most reliable lever governments have for reducing household OOP burden, with elasticity an order of magnitude larger than that of income growth. Inflation, often invoked in policy discussions as a driver of household healthcare strain, has no detectable effect on the OOP share in the data, though it doubtless affects the absolute level of OOP spending in ways the share specification masks.

6. Conclusion

Using a balanced panel of 188 countries from 2000 to 2024 and a country fixed-effects specification, this paper finds that the share of out-of-pocket health expenditure in current health spending is robustly and negatively associated with the share of government health expenditure and with the log of GDP per capita. Population ageing exerts a smaller positive effect, and inflation has no detectable effect. The within-country evidence is consistent with the policy intuition that expanding public financing displaces direct household payment for health, and quantifies that displacement at roughly 4.75 percentage points of the OOP share per percentage point of the public share. Further work should pursue dynamic specifications that explicitly model the persistence of OOP shares and disaggregate by World Bank income group to test whether the substitution mechanism operates similarly at every stage of development.

[11] Ke, Saksena, and Holly, 'Determinants of Health Expenditure', p. 16.

[12] Kaladharan and Manayath, 'Out-of-Pocket Healthcare Expenditure', p. 8.

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